# ALEXANDER TONG

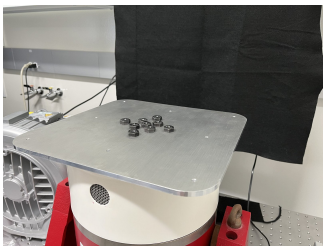
Mechanical engineering | Recovery hardware and supersonic aerodynamics

Boston University, B.S. Mechanical Engineering, Aerospace Concentration | Expected May 2028
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**PORTFOLIO AT A GLANCE**

## Project Index

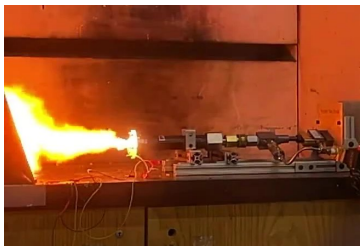
| | |
|---|---|
| Page 2 / Separation & recovery | Earlier black-powder configuration: 18.7 lbm within a 20 lbm allocation. Load sizing, CO2 redesign and an operational vacuum chamber. |
| Page 3 / Aerodynamic fairing | Three concept studies; Ansys CFD baseline converged and under review. Shared aerodynamic analysis, integration and manufacturing trade. |
| Page 4 / Plummer shaker hardware | Structural analysis, bored-bolt checks and CNC fabrication. Installed plate-and-bolt hardware at 93.9% lower estimated cost. |



Plummer: machined adapter installed.



AIAA: assembled Prototype 1 aircraft.



Hybrid nozzle: hotfire test frame.

### Additional Projects

**Plummer Laboratory**

CNC-machined shaker adapter; plate-and-bolt hardware estimate reduced from $1,170 to $71.11, excluding labor. Designed the matrix and procedure for 48 Instron tests.

**AIAA Design-Build-Fly wing**

Complete Prototype 1 wing 42.7% lighter than the previous complete wing; build time reduced from three months to one. Led design, fabrication and member training.

**Hybrid rocket nozzle**

73% mass reduction from my initial design; approximately two-second hotfire. Minimum yield FOS 7.8 in SolidWorks FEA for mechanical loads only; thermal effects excluded. Thrust gain unmeasured.

**EK210 line-following car**

Designed and programmed the vehicle: six loaded course runs in 70-86 seconds against a 90-second limit. Recorded component cost $73.94 against a $200 budget.

**Culturon**

Executed the company's first full-scale nanoparticle deposition map using microscopy and ImageJ. Designed adhesive-free sample racks; presented prevention concepts.

**Manufacturing & mentoring**

CNC and manual machining, hot-wire foam cutting and composite fabrication. Taught BURPG members turning and milling through the machinist's-hammer project beginning Fall 2025.

### Additional research & coursework

Space Physics & Technology Lab: instrument-data analysis with Python/pySPEDAS and assistance with flight-spare laser calibration. University of Sydney pulsed plasma thruster: inductor-spool design and coil-winding calculations. ME360 course project includes completed linear-stage/CNC drawing and fishing-game builds.

01 / RECOVERY HARDWARE

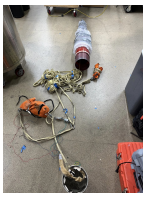
# Argo Separation and Recovery Subsystem

Boston University Rocket Propulsion Group | Subsystem Co-Lead | June 2025-present

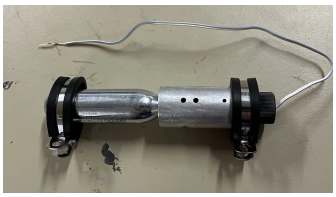
**18.7 / 20 lbm**
Earlier BP mass / allocation; BP BOM $1,076

**823 / 1,230 lbf**
Opening estimate / initial line-sizing case

**352 L**
Operational vacuum chamber; reported test pressure 0.011 atm



Subsystem before CO2 coldflow; historical pretest photo.



Manufactured CO2 cartridge/piercer assembly. Design and fabrication shared by our three-person team.



Door, adapter and seal designed by me; fabricated by my co-lead, who designed the clamps.

## Design and Test Leadership

Developed hardware within inherited single-break, dual-deploy recovery through four reviews. Compared five load methods; applied a 1.5 comparison allowance to the 823 lbf estimate for initial sizing. Paired flight computers provide separate firing paths.

Authored the integrated vinyl-charge black-powder SOP and led the second integrated test; collaborated on inspection and failure assessment. One of three charges failed in each test; the second damaged the parachute bag and canopy.

Standalone successes did not establish integrated reliability. End-seal variability is the leading hypothesis; adjacent-charge damage was inferred. Root cause remains unconfirmed.

Designed and sized CO2 piercers using pressure, Lamé thick-wall hoop-stress and thread-shear calculations; contributed CAD, drawings, machining and component testing in a three-person team.

## Calculation basis

| | |
|---|---|
| Recovery lines | Selected 1/2 in, 6,000 lbf Kevlar stock. Design rationale: near-full strength retention, approximately 2× a 50%-efficient knot. Our strength gain remains unmeasured. |
| CO2 deployment sizing | Calculated 4.66 psi separation requirement for four nylon shear pins plus estimated interface friction. Ideal-gas estimate for the modeled free volume: 10.9 g versus a selected 16 g cartridge. |
| CO2 piercer housing | Nominal hoop ratio 2.01; thread shear 6.31. Saved combined stress gives ~29.2 ksi von Mises, ratio 1.37 using 40 ksi typical yield. Material/pressure basis and local stresses remain unverified. |
| Vacuum door / Roark | 12 in unsupported diameter, 0.25 in thick 6061-T6 plate; 14.7 psi differential gives 1,663 lbf total load. Recorded bending stress 16.5 ksi; deflection 0.0848 in; minimum yield FOS 2.12 using 240 MPa yield strength. Uniform pressure; simply supported around the outer circumference. Deflection uses the Roark plate model; stress is a separately entered calculator result. |
| Legacy Catclaw body | Weighed mass reduction: 49.5%; piston-retention slot revised. Later Lamé ratio reported as 2.01; source and yield criterion under review. Separate from the CO2 piercer. |

## Verification Status

Component firing completed at reported 0.011 atm initial pressure. The integrated cartridge discharged, but the nosecone did not separate; a wiring-port leak prompted an NPT/epoxy/putty seal revision. Retest and flight testing are pending. Splice trial: no observed slip; earlier 1,800 lbf interval-average estimate awaits timing/unit checks. Peak force and ultimate strength unmeasured. Chamber instrumentation and pressure analysis by my co-lead.

Recovery architecture, hardware & tests | CO2 sizing & stress calculations | Chamber design

02 / AERODYNAMICS

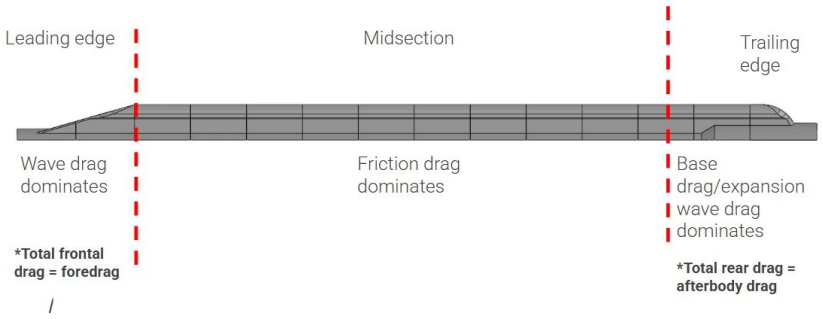
# Argo Downcomer Aerocover

Boston University Rocket Propulsion Group | Project Lead | August 2026-present

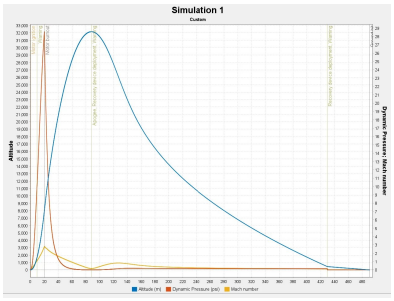
**63 in × 1.6 in**
Fairing length / nominal height on an 8 in diameter rocket

**Mach 2.0-2.81**
Priority comparison range from the simulated trajectory

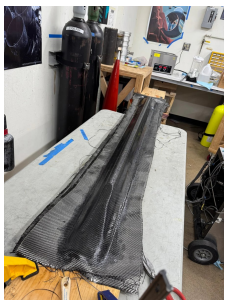
**3 concepts**
Teammate-owned geometries; shared requirements and analysis by me




My CoDR diagram: a first-principles decomposition, not a measured allocation of drag. Pressure and friction act across the whole fairing.




Trajectory used to select high-dynamic-pressure analysis conditions; simulation, not flight data.



Legacy Icarus fairing: the CFD baseline and a backup flight candidate, not a new design by me.

## Analysis

Used D = q Sref CD to prioritize conditions by dynamic pressure as well as Mach. Evaluated leading-edge pressure drag, midsection wetted area and aft pressure recovery together with thermal loads, attachment integrity and manufacturing constraints. Isolated axisymmetric profile rankings cannot establish the performance of a partial fairing attached to the rocket body: junction flow and fin interference require integrated analysis.

| | |
|---|---|
| Preliminary friction estimate | At Mach 2.81 and recorded Re = 2.04 × $10^8$: $C_{D,f} \approx 0.0792$, using 477 in² wetted area, 7.99 in² reference area and a 10% preliminary roughness allowance. |
| Leading-edge surrogate | A 15° full-cone correlation gives $C_p \approx 0.1888$. It is a screening surrogate for the wedge-like legacy front, not the mounted fairing's total drag coefficient. |

## Contributions and Status

I developed the common analysis, integration/manufacturing trade and CoDR. Quinn Colleran, Hannah Ruan and Aaron Lin own the three concept geometries. My Mach 2.81 Icarus solve in **Ansys Fluent is converged and under review**; candidate ranking and Ansys adjoint optimization remain pending.

The investigation also led to a nine-profile-family nosecone sweep in RASAero II. A ranking difference from published comparisons prompted further investigation; it does not establish a better flight design.

Concept credits, hand calculations, sources & nosecone sweep | Hand-calculation reference: S. S. Chin, Missile Configuration Design (1961).

03 / BU PLUMMER LABORATORY

# Shaker Adapter and Mounting Bolts

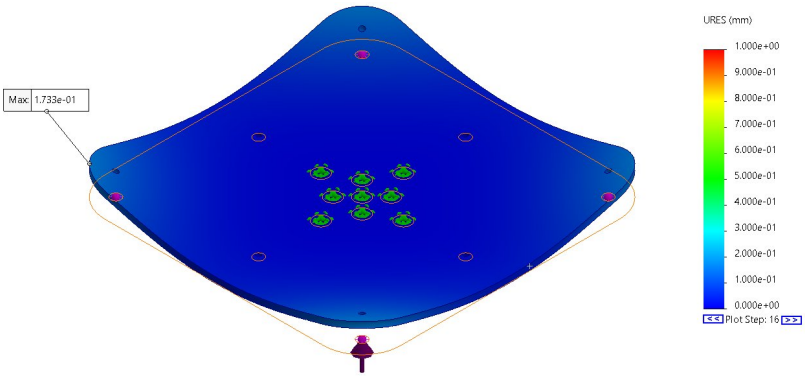
Mechanical Engineering Intern | June 2026-present

**0.173 mm**
Plate displacement predicted in the design-review FEA case

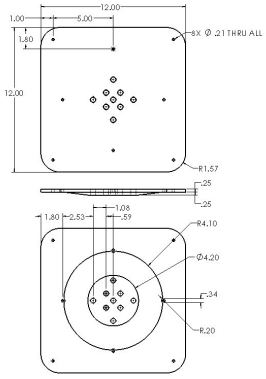
**20.1 Hz**
Shaker operating frequency; analyzed with a 3 kg fixture load

**2.46 FOS**
Plate bearing under the modeled combined bolt load

The adapter mounts a physical Ising-model fixture for controlled shaker vibration tests of bistable buckling elements.




SolidWorks plate-displacement result: 0.173 mm maximum. This is a model prediction.




My 12 × 12 in adapter drawing, including underside geometry.

## Design and Manufacturing

Replaced an estimated $1,170 outsourced plate/custom-bolt solution with $71.11 of in-house hardware. Developed SolidWorks CAD/FEA, drawings and CAM/G-code; machined the 6061-T6 plate on a Haas VF-2 and checked plate dimensions with calipers against the ±0.005 in drawing tolerance.

Load-path and displacement calculations guided material removal; SolidWorks FEA checked the lightened plate. Finished mass: 1.90 kg from 3.19 kg stock (40.4% stock-to-part reduction).

Drilled 5 mm axial bores in standard medium-carbon-steel M10 bolts for the required shaker interface after unsuccessful stainless-steel machining attempts. The bored bolts and adapter are installed.

## Structural calculation basis

| | |
|---|---|
| Installed joint geometry | M10 × 1.5 with a 5 mm axial bore; modeled remaining tensile area 38.35 mm². Nominal Class 8.8 yield strength 640 MPa; local plate bearing thickness 9.7 mm; thread engagement 9.35 mm. |
| Per-bolt load path | 5 kg allowance (2 kg plate + 3 kg fixture) at 11 G, shared by four bolts: 134.9 N external load per bolt. Workbook conservatively adds the full external load to 16.36 kN preload: 16.50 kN combined. Torque estimate: 21.7 ft-lbf at K = 0.18; joint stiffness sharing is not modeled. |
| Static FOS by checked mode | Bolt tensile yield: 1.49 under preload plus full external load; preload alone gives 1.50. Plate bearing 2.46; pull-through 4.99; external-thread shear 3.71; internal-thread shear 3.60. These are mode-specific ratios, not an overall dynamic FOS. |
| Displacement checks | Selected L/240 limit for the 12 in plate span: 1.27 mm; review plate FEA: 0.173 mm. The later 3 kg dynamic fixture model predicts 0.44 mm deformation relative to the base at 20.1 Hz and 13 mm peak-to-peak base motion. These are separate load cases. |

Preload maintains joint compression; bolt/joint stiffness determines cyclic load sharing. The 1.49 ratio checks static tensile yield, not fatigue at 20.1 Hz. Preload scatter, separation and fatigue are not quantified here. Displacements are predictions; cost covers plate and bolts only.

Full calculations, drawings, FEA and machining evidence | Sources: my design review and DSRF Calcs; joint interpretation: NASA TM-106943.